

Cyclolux

0.5 mmol/ml solution for injection

Pharmacode 827

204827/00

1. NAME OF THE MEDICINAL PRODUCT

CYCLOLUX

0.5 mmol/ml solution for injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1 ml solution for injection contains 279.32 mg gadoteric acid (as meglumine salt) corresponding to Gadolinium (III) oxide 90.62mg, DOTA 202.21mg, Meglumine 97.6mg.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Gadoteric acid (as meglumine salt) is a clear, colourless to pale yellowish solution, free from visible particulates.

Contrast medium concentration	279.32 mg/ml 0.5 mmol/ml
Osmolality at 37°C	1.35 Osm/kg H ₂ O
Viscosity at 37°C	1.8 mPas
pH value	6.5 – 8.0

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

This medicinal product is for diagnostic use only.

Cyclolux should be used only when diagnostic information is essential and not available with unenhanced magnetic resonance imaging (MRI).

Magnetic resonance imaging for:

- Cerebral and spinal disease
- Disease of the vertebral column
- and other whole-body pathologies (including Angiography)

4.2 Posology and method of administration

Posology

The lowest dose that provides sufficient enhancement for diagnostic purposes should be used. The dose should be calculated based on the patient's body weight, and should not exceed the recommended dose per kilogram of body weight detailed in this section.

Adults
Cyclolux should always be administered by medically qualified personnel.
The doses are determined by the physician and are adapted to each individual's needs.
The recommended dose is 0.2ml/kg (0.1mmol/kg) in adult, children and infant.
In angiography, depending on the result of examination being performed, a second injection may be administered during the same session if necessary.
In some exceptional cases, as in the confirmation of isolated metastasis or the detection of leptomeningeal tumours, a second injection of 0.4ml/kg (0.2mmol/kg) can be administered.

Special populations
Impaired renal function
The adult dose applies to patients with mild to moderate renal impairment (GFR ≥ 30 ml/min/1.73m²).

Cyclolux should only be used in patients with severe renal impairment (GFR < 30 ml/min/1.73m²) and in patients in the perioperative liver transplantation period after careful risk/benefit assessment and if the diagnostic information is essential and not available with non-contrast enhanced MRI (see section "Special warnings and precautions for use"). If it is necessary to use Cyclolux, the dose should not exceed 0.1 mmol/kg body weight. More than one dose should not be used during a scan. Because of the lack of information on repeated administration, Cyclolux injections should not be repeated unless the interval between injections is at least 7 days.

Elderly (aged 65 years and above)
No dosage adjustment is considered necessary. Caution should be exercised in elderly patients.

Impaired hepatic function
The adult dose applies to these patients. Caution is recommended, especially in the case of perioperative liver transplantation period.

Paediatric population (0-18 years)
MRI of brain and spine / whole-body MRI: the recommended and maximum dose of Gadoteric acid is 0.1 mmol/kg body weight. More than one dose should not be used during a scan.
Due to immature renal function in neonates up to 4 weeks of age and infants up to 1 year of age, Cyclolux should only be used in these patients after careful consideration at a dose not exceeding 0.1 mmol/kg body weight. Because of the lack of information on repeated administration, Cyclolux injections should not be repeated unless the interval between injections is at least 7 days.
Angiography: Cyclolux is not recommended for angiography in children under 18 years of age due to insufficient data on efficacy and safety in this indication.

Method of administration
This product must be administered by intravenous injection only.

Intravascular administration of contrast media should, if possible, be done with the patient lying down. After the administration, the patient should be kept under observation for at least half an hour, since experience shows that the majority of undesirable effects occur within this time.

Paediatric population (0-18 years)
Depending on the amount of gadoteric acid to be given to the child, it is preferable to use gadoteric acid vials with a single use syringe of a volume adapted to this amount in order to have a better precision of the injected volume.
In neonates and infants, the required dose should be administered by hand.
This product is for single-use.

4.3 Contraindications
Hypersensitivity to gadoteric acid, to meglumine or to any medicinal products containing gadolinium.

4.4 Special warnings and precautions for use
Do not use by intrathecal route. Take care to maintain strictly intravenous injection: extravasation may result in local intolerance reactions, requiring the usual local care.

The usual precaution measures for MRI examination should be taken, such as exclusion of patients with pacemakers, ferromagnetic vascular clips, infusion pumps, nerve stimulators, cochlear implants, or suspected intracorporal metallic foreign bodies, particularly in the eye.

Hypersensitivity
- As with other gadolinium containing contrast media hypersensitivity reactions can occur, including life-threatening (see section undesirable effects). Hypersensitivity reactions may be either allergic (described as anaphylactic reactions when serious) or non-allergic. They can be either immediate (less than 60 minutes), or delayed (up to 7 days). Anaphylactic reactions occur immediately and can be fatal. They are independent of the dose, can occur after even the first dose of the product, and are often unpredictable.

- There is always a risk of hypersensitivity regardless of the dose injected.
- Patients who have already experienced a reaction during previous administration of a gadolinium-containing MRI contrast agent present an increased risk of experiencing another reaction on subsequent administration of the same product, or possibly other products, and are therefore considered to be at high risk.
- The injection of gadoteric acid may aggravate symptoms of an existing asthma. In patients with asthma unbalanced by the treatment, the decision to use gadoteric acid must be made after careful evaluation of the risk/benefit ratio.
- As known from the use of iodinated contrast media, hypersensitivity reactions can be aggravated in patients on beta-blockers, and particularly in the presence of bronchial asthma. These patients may be refractory to standard treatment of hypersensitivity reactions with beta-agonists.

- Before any contrast medium is injected, the patient should be questioned for a history of allergy (e.g., seafood allergy, hay fever, hives), sensitivity to contrast media and bronchial asthma as the reported incidence of adverse reactions to contrast media is higher in patients with these conditions and premedication with antihistamines and/or glucocorticoids may be considered.
- During the examination, supervision by a physician is necessary. If hypersensitivity reactions occur, administration of the contrast medium must be discontinued immediately and - if necessary - specific therapy instituted. A venous access should thus be kept during the entire examination. To permit immediate emergency countermeasures, appropriate drugs (e.g., epinephrine and antihistamines), an endotracheal tube and a respirator should be ready at hand.

Impaired renal function
Prior to administration of Cyclolux, it is recommended that all patients are screened for renal dysfunction by obtaining laboratory tests.

There have been reports of nephrogenic systemic fibrosis (NSF) associated with use of some gadolinium- containing contrast agents in patients with acute or chronic severe renal impairment (GFR < 30 ml/min/1.73m²). Patients undergoing liver transplantation are at particular risk since the incidence of acute renal failure is high in this group. As there is a possibility that NSF may occur with Cyclolux, it should therefore only be used in patients with severe renal impairment and in patients in the perioperative liver transplantation period after careful risk/benefit assessment and if the diagnostic information is essential and not available with non-contrast enhanced MRI.

Haemodialysis shortly after gadoteric acid administration may be useful at removing gadoteric acid from the body. There is no evidence to support the initiation of haemodialysis for prevention or treatment of NSF in patients not already undergoing haemodialysis.

Elderly
As the renal clearance of gadoteric acid may be impaired in the elderly, it is particularly impor-

tant to screen patients aged 65 years and older for renal dysfunction.

Paediatric population
Neonates and infants
Due to immature renal function in neonates up to 4 weeks of age and infants up to 1 year of age, Cyclolux should only be used in these patients after careful consideration.

Cardiovascular disease
In patients with severe cardiovascular disease Cyclolux should only be administrated after careful risk benefit assessment because only limited data are available so far.

CNS disorders
Like with other gadolinium containing contrast agents special precaution is necessary in patients with a low threshold for seizures. Precautionary measures should be taken, e.g. close monitoring. All equipment and drugs necessary to counter any convulsions which may occur must be made ready for use beforehand.

- Exposure to gadolinium – based contrast agents (GBCAs) increases the risk for Nephrogenic Systemic Fibrosis (NSF) in patients with:
 - acute or chronic severe renal insufficiency (glomerular filtration rate < 30mL/min/1.73m²), or
 - acute renal insufficiency of any severity due to the hepato-renal syndrome or in the perioperative liver transplantation period.
- NSF is a debilitating and sometimes fatal disease affecting the skin, muscle, and internal organs
- Avoid use of GBCAs unless the diagnostic information is essential and not available with non-contrast enhanced magnetic resonance imaging (MRI).
- Screen all patients for renal dysfunction by obtaining a history and/ or laboratory tests.
- When administering a GBCA, do not exceed the dose recommended in product labelling.

Allow sufficient time for elimination of the GBCA prior to any readministration.

WARNING

- Among the factors that may increase the risk for NSF are repeated or higher than recommended doses of a GBCA.
- For patients receiving haemodialysis, healthcare professionals may consider prompt haemodialysis following GBCA administration in order to enhance the contrast agent's elimination. However, it is unknown if haemodialysis prevents NSF.
- Determine the renal function of patients by obtaining a medical history of conducting laboratory tests that measure renal function prior to using GBCA.
- The risk, if any, for developing NSF among patients with mild to moderate renal insufficiency or normal renal function is unknown.
- Post marketing reports have identified the development of NSF following single and multiple administrations of GBCAs.

4.5 Interaction with other medicinal products and other forms of interaction

No interactions with other medicinal products have been observed. Formal drug interaction studies have not been carried out.

Concomitant medications to be taken into account
Beta-blockers, vasoactive substances, angiotensin-converting enzyme inhibitors, angiotensin II receptor antagonists: these medicinal products decrease the efficacy of the mechanisms of cardiovascular compensation for blood pressure disorders: the radiologist must be informed before injection of gadolinium complexes, and resuscitation equipment must be at hand.

4.6 Fertility, pregnancy and lactation
Pregnancy

There are no data from the use of gadoteric acid in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. Cyclolux should not be used during pregnancy unless the clinical condition of the woman requires use of gadoteric acid.

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Fontsize: 10 Pt
Colour: Pantone 280 C

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Lactation

Gadolinium containing contrast agents are excreted into breast milk in very small amounts. At clinical doses, no effects on the infant are anticipated due to the small amount excreted in milk and poor absorption from the gut. Continuing or discontinuing breast feeding for a period of 24 hours after administration of Cyclolux, should be at the discretion of the doctor and lactating mother.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Ambulant patients while driving vehicles or operating machinery should take into account that nausea may incidentally occur.

4.8 Undesirable effects

Side effects in association with the use of gadoteric acid are usually mild to moderate in intensity and transient in nature. Injection site reactions, nausea and headache are the most frequently observed reactions. During clinical trials, nausea, headache, injection site reactions, feeling cold, hypotension, somnolence, dizziness, feeling hot, burning sensation, rash, asthenia, dysgeusia and hypertension were the most frequent, uncommonly observed (≥1/1000 to <1/100) related adverse events.

Since post-marketing, the most commonly reported adverse reactions following administration of gadoteric acid are nausea, vomiting, pruritus and hypersensitivity reactions.

In hypersensitivity reactions, the reactions most frequently observed are skin reactions, which can be localized, extended or generalized.

These reactions occur most often immediately (during the injection or within one hour after the start of injection) or sometimes delayed (one hour to several days after injection), presenting as skin reactions in this case.

Immediate reactions include one or more effects, which appear simultaneously or

sequentially, which are most often cutaneous, respiratory, gastrointestinal, articular and/or cardiovascular reactions.

Each sign may be a warning sign of a starting shock, however, it is very rarely fatal.

Isolated cases of nephrogenic systemic fibrosis (NSF) have been reported with gadoteric acid, most of which were in patients co-administered other gadolinium-containing contrast agents (see section Interaction with other medicinal products and other forms of interactions).

The adverse reactions are listed in the table below by SOC (System Organ Class) and by frequency with the following guidelines: very common, common, uncommon, rare, very rare, not known.

System Organ Class	Frequency:	adverse reaction
Immune system disorders	Uncommon: Very rare:	hypersensitivity anaphylactic reaction, anaphylactoid reaction
Psychiatric disorders	Rare: Very rare:	anxiety agitation
Nervous system disorders	Uncommon: Rare: Very rare:	headache, dysgeusia, dizziness, somnolence, paraesthesia (including burning sensation) presyncope coma, convulsion, syncope, tremor, parosmia
Eye disorders	Rare: Very rare:	eyelid edema conjunctivitis, ocular hyperaemia, vision blurred, lacrimation increased
Cardiac disorders	Rare: Very rare:	palpitations tachycardia, cardiac arrest, arrhythmia, bradycardia
Vascular disorders	Uncommon: Very rare:	hypotension, hypertension pallor, vasodilatation
Respiratory, thoracic and mediastinal disorders	Rare: Very rare:	sneezing cough, dyspnoea, nasal congestion, respiratory arrest, bronchospasm, laryngospasm, pharyngeal oedema, dry throat, pulmonary oedema
Gastrointestinal disorders	Uncommon: Rare:	nausea, abdominal pain vomiting, diarrhoea, salivary hypersecretion
Skin and subcutaneous tissue disorders	Uncommon: Rare: Very rare: Not known:	rash urticaria, pruritus, hyperhidrosis erythema, angioedema, eczema nephrogenic systemic fibrosis
Musculoskeletal and connective tissue disorders	Very rare:	muscle cramps, muscular weakness, back pain
General disorders and administration site conditions	Uncommon: Rare: Very rare:	feeling hot, feeling cold, asthenia, injection site reactions (extravasation, pain, discomfort, oedema, inflammation, coldness) chest pain, chills malaise, chest discomfort, pyrexia, face oedema, injection site necrosis (in case of extravasation), phlebitis superficial
Investigations	Very rare:	decreased oxygen saturation

The following adverse reactions were reported with other intravenous contrast agents for MRI:

Organ Class System	Adverse reaction
Blood and lymphatic system disorders	Haemolysis
Psychiatric disorders	Confusion
Eye disorders	Blindness transient, eye pain
Ear and labyrinth disorders	Tinnitus, ear pain
Respiratory, thoracic and mediastinal disorders	Asthma
Gastrointestinal disorders	Dry mouth
Skin and subcutaneous tissue disorders	Dermatitis bullous
Renal and urinary disorders	Urinary incontinence, renal tubular necrosis, renal failure acute
Investigations	Electrocardiogram PR prolongation, blood iron increased, blood bilirubin increased, serum ferritin increased, liver function test abnormal

Adverse reaction in Children

Safety of paediatric patients was considered in clinical trials and post marketing studies. As compared to adult, the safety profile of gadoteric acid did not show any specificity in children. Most of reactions are gastrointestinal symptoms or signs of hypersensitivity.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions to local Health Authority.

4.9 Overdose

Gadoteric acid can be removed by haemodialysis. However there is no evidence that haemodialysis is suitable for prevention of nephrogenic systemic fibrosis (NSF).

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: paramagnetic contrast media for MRI
ATC code: V08CA02 (gadoteric acid).

Cyclolux has paramagnetic properties allowing MRI contrast enhancement. It has no specific pharmacodynamic activity and is biologically very inert.

5.2 Pharmacokinetic properties

Following intravenous injection, gadoteric acid is mainly distributed in the extracellular fluid. It is not bound to plasma albumin. In patients with normal renal function, the plasma half-life is about 90 minutes. Gadoteric acid is eliminated in unchanged form by glomerular filtration. Plasma clearance is delayed in patients with impaired renal function. A small amount of gadoteric acid is excreted in breast milk and crosses the placenta.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Meglumine, DOTA, Water for injections

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

3 years.

6.4 Special precautions for storage

Store below 30 °C.

6.5 Nature and contents of container

1 and 10 Type II single-use colourless glass vials containing 10mL, 15mL or 20mL, sealed with a stopper of bromobutyl rubber and packed in unit carton box.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

The peel-off tracking label on the vials should be stuck onto the patient record to enable accurate recording of the gadolinium contrast agent used. The dose used should also be recorded. If electronic patient records are used, the name of the product, the batch number and the dose should be entered into the patient record.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

6.7 Instruction of Use

This product must be administered by intravenous injection only.

Optimal imaging: within 45 minutes after injection

Optimal image sequence: T1-weighted
Intravascular administration of contrast media should, if possible, be done with the patient lying down. After the administration, the patient should be kept under observation for at least half an hour, since experi-

ence shows that the majority of undesirable effects occur within this time.

Prepare a syringe with a needle. Remove the plastic disk. After cleaning the stopper with a pad soaked in alcohol, puncture the stopper with the needle. Withdraw the quantity of product required for the examination and inject it intravenously.

For single use only, any unused solution should be discarded.

The solution for injection should be inspected visually prior to use. Only clear solutions free of visible particles should be used.

7. MARKETING AUTHORISATION HOLDER

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Selangor, MALAYSIA

8. MANUFACTURER

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9. DATE OF REVISION OF THE TEXT

21/07/2022